

Computer Networking: Concepts Experiment 6

PART 1 — FIXED-LENGTH SUBNETTING

Given Network: **172.16.0.0/24**

Required Subnets:

- Subnet A: 40 hosts
- Subnet B: 20 hosts
- Subnet C: 10 hosts
- Subnet D: 5 hosts

! *Fixed-length subnetting means all subnets must be of equal size.*

Step 1 — Determine Subnet Size

We need the largest host requirement → **40 hosts**.

Formula:

$$\text{Hosts} = 2^h - 2 \geq 40$$

$$2^6 - 2 = 62 \rightarrow \text{OK}$$

So we need **6 host bits**, leaving:

$$\text{Prefix} = 32 - 6 = /26$$

$$\text{Subnet mask} = 255.255.255.192$$

Each subnet size = $2^6 = 64$ addresses per subnet

Step 2 — Create Subnets of /26

Subnet	Network Address	First Host	Last Host	Broadcast
A	172.16.0.0/26	172.16.0.1	172.16.0.62	172.16.0.63
B	172.16.0.64/26	172.16.0.65	172.16.0.126	172.16.0.127
C	172.16.0.128/26	172.16.0.129	172.16.0.190	172.16.0.191
D	172.16.0.192/26	172.16.0.193	172.16.0.254	172.16.0.255

Step 3 — Packet Tracer Topology

Use:

- 1 Router
- 4 Switches
- Multiple PCs in each subnet

To add new interfaces (Extra Ports)

Drag the Router to the Workspace

- From **Devices** → **Routers**
- Choose **Cisco 2911**
- Drag onto the workspace.

Power OFF the Router

You cannot insert modules while powered on.

- Click the router
- Go to the **Physical** tab
- Find the **Power button** (left side → switch)
- Click it → It turns **red (OFF)**

Choose a Module to Add

Common modules for extra ports:

Module Name	Ports Added	Purpose
HWIC-2T	2 Serial ports	WAN/Serial links

<i>Module Name</i>	<i>Ports Added</i>	<i>Purpose</i>
HWIC-4ESW	4 FastEthernet switch ports	LAN expansion
HWIC-1GE-SFP	1 GigE	High-speed port
NIM-ES2-4	4 Ethernet ports	Newer PT versions

For subnetting labs, the **HWIC-4ESW** is the easiest and most used.

Insert the Module

- In the router's Physical tab, you'll see **empty HWIC slots**.
- Drag the module (e.g., **HWIC-4ESW**) from the left panel
- Drop it into an empty slot.

You will see the module appear with new ports like:

Fa0/0
Fa0/1
Fa0/2
Fa0/3

Power ON the Router

- Click the same power switch
- It turns **green (ON)**

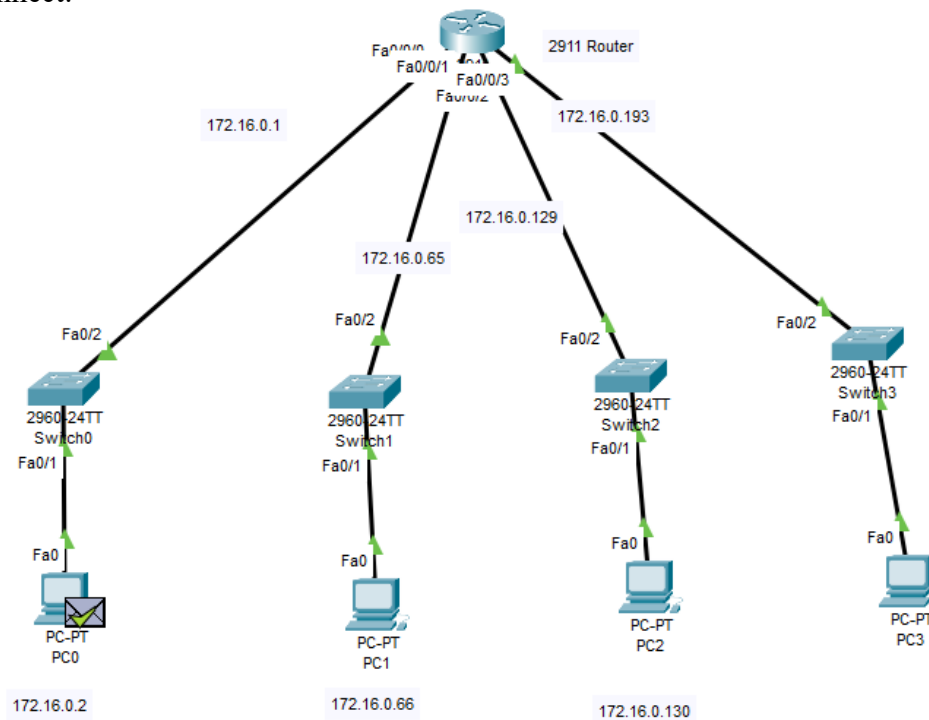
Go to CLI and Verify Ports

Open CLI → type:

`show ip interface brief`

You should now see the new interfaces.

Connect:



Step 4 — Router Configuration

Example for Router:

`enable`

`configure terminal`

`interface g0/0`

`ip address 172.16.0.1 255.255.255.192`

`no shutdown`

```
interface g0/1
  ip address 172.16.0.65 255.255.255.192
  no shutdown

interface g0/2
  ip address 172.16.0.129 255.255.255.192
  no shutdown

interface g0/3
  ip address 172.16.0.193 255.255.255.192
  no shutdown
```

Since a single router is used, **routing is automatic** (no static routes needed).

Step 5 — PC Configuration Example

PC in Subnet A:

IP: 172.16.0.10

Mask: 255.255.255.192

Gateway: 172.16.0.1

Repeat for all PCs.

Step 6 — Verification

Ping across subnets:

PC-A → PC-B

PC-A → PC-C

PC-A → PC-D

All pings should succeed.

PART 2 — VLSM

We use the same network **172.16.0.0/24**, but assign subnet sizes based on host counts.

Host requirements:

- A → 40
- B → 20
- C → 10
- D → 5

Step 1 — Sort by host size

1. A = 40 hosts → needs /26 (62 hosts)
2. B = 20 hosts → /27 (30 hosts)
3. C = 10 hosts → /28 (14 hosts)
4. D = 5 hosts → /29 (6 hosts)

Step 2 — Allocate Subnets (largest first)

✓ **Subnet A (40 hosts) → /26**

172.16.0.0/26

Range: 172.16.0.1 - 172.16.0.62

Broadcast: 172.16.0.63

✓ **Subnet B (20 hosts) → /27**

Next available block starts at 172.16.0.64

172.16.0.64/27

Range: 172.16.0.65 - 172.16.0.94

Broadcast: 172.16.0.95

✓ **Subnet C (10 hosts) → /28**

Next block starts at 172.16.0.96

172.16.0.96/28

Range: 172.16.0.97 - 172.16.0.110

Broadcast: 172.16.0.111

✓ **Subnet D (5 hosts) → /29**

Next block starts at 172.16.0.112

172.16.0.112/29

Range: 172.16.0.113 - 172.16.0.118

Broadcast: 172.16.0.119

Step 3 — Router Interface Assignment

Do same as previous

Step 4 — Configure PCs

Example for Subnet C:

IP: 172.16.0.100

Mask: 255.255.255.240

Gateway: 172.16.0.97

Step 5 — Verify With Ping

All hosts must successfully ping across all networks.